bankpulse-term-deposit-v1

Model card details

Model card version

1

Model card status

Draft

Created date

2026-10-20T19:47:36Z

KMS encryption key

-

Model card ARN

arn:aws:sagemaker:us-east-1:973712293028:model-card/bankpulse-term-deposit-v1

Model details

Associated model

This model card is not associated with a model group, hosted, or external model.

General information

Model description

Gradient-boosted trees (built-in SageMaker XGBoost 1.7-1) scoring the probability that a customer subscribes to a term deposit after a phone call.

Model version

0

Problem type

Binary Classification

Algorithm type

XGBoost

Model creator

-

Model owner

bankpulse-ds

Business details

Business problem

-

Business stakeholders

-

Line of business

-

Intended uses

Intended uses of the model

Prioritising call lists for term deposit campaigns, reviewed by campaign staff. Not for credit, lending, pricing or any decision about eligibility for a product.

Purpose of the model

Rank customers before a telemarketing campaign so the most likely subscribers are called first.

Factors affecting model efficiency

Economic conditions (Euribor, employment) change over time and drive subscription rates; scores degrade when they drift.

Risk rating

Medium

Explanations for risk rating

Affects who is contacted, not access to products; uses age and job, so selection rates by age band are monitored.

Training details

Objective function

Objective function

Notes on objective function

-

Training observations

Trained on the oldest 70% of calls (time-ordered split), validated on the next 15%. Call duration is excluded because it is only known after the call (leakage).

Training job details

Training job ARN

arn:aws:sagemaker:us-east-1:973712293028:training-job/bankpulse-xgboost-2026-10-20-19-38-10-728

Input dataset used to train the model

-

Training environment

-

Training results

Results from training job

User entered training results

Hyperparameters

User entered hyperparameters

Evaluation details

Validation (15%, after training period)

Evaluation observations

Decision threshold 0.1064 chosen here by best F1.

Evaluation datasets

-

Additional evaluation metadata

Evaluation results

binary classification

Metric	Value	User notes
pr_auc	0.2283	

roc_auc	0.6619	
precision	0.2604	
recall	0.3419	
f1	0.2957	
lift_top_decile	2.663	
positive_rate	0.1065	

Test (newest 15%)

Evaluation observations

Evaluated once. Subscription rate is 38% vs 6% in training: strong drift, so absolute scores are not comparable across periods.

Evaluation datasets

-

Additional evaluation metadata

Evaluation results

binary classification

Metric	Value	User notes
pr_auc	0.4673	
roc_auc	0.6291	
precision	0.4733	
recall	0.3022	
f1	0.3689	
lift_top_decile	1.239	
positive_rate	0.3845	

Additional details

Ethnical considerations

Age and job are inputs. Selection rates per age band are reported by the monitoring notebook (05); a min/max ratio below 0.8 triggers review.

Caveats and recommendations

Retrain when monitoring shows feature PSI above 0.2 on the economic indicators. Data from one Portuguese bank, 2008-2010.

Custom details

model_package_arn

arn:aws:sagemaker:us-east-1:973712293028:model-package/bankpulse-term-deposit/1

baseline_test_pr_auc

0.4712